



Preparing for the Next Pandemic: Learning Lessons from the Recent Past

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Abstract

While the initial waves of the COVID-19 pandemic have long come to a close, the pandemic presents an opportunity to learn from the recent past in order to prepare for the next one. Throughout the COVID-19 pandemic, vaccination rates were high in some parts of the world and stalled in others as new variants emerged. Non-pharmaceutical interventions (NPIs) played an important role in limiting COVID-19 spread. But, how important of a role? Using a Susceptible-Exposed-Asymptomatic-Infected-Recovered (SEAIR) model, the proportion of

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the population that followed NPI guidelines (e.g., wearing a mask) were varied while changing the ratio of vaccinations that were given to individuals that follow NPI guidelines to those individuals that do not follow established guidelines. For each of the scenarios, the peak number of infections and overall number of infections during the outbreak were tracked. Both metrics can be considered gauges of hospital and societal burden associated with the disease. The simulations show NPIs are effective at controlling outbreaks. However, when relatively few individuals comply with NPI guidelines, the most effective strategy to decrease burden becomes prioritizing the vaccination of the non-compliers – those not conforming to NPI guidelines. With that in mind, vaccination campaigns focused on specific groups such as individuals not following suggested NPI guidelines may be the best way to increase vaccination uptake, help stop the spread of the disease, and decrease overall societal burden. The results lend insight into potential mitigation strategies that should be considered during the next pandemic, which is surely closer than one thinks.

Keywords

SEAIR model · Pandemic · COVID-19 · Stochastic dynamics · Hospital burden · Societal burden

1 Introduction

While the initial waves of COVID-19 have long come to a close around the world, the next pandemic, unfortunately, may be around the corner. Given that, we need to recognize that non-pharmaceutical interventions or NPIs (e.g., wearing a mask in public) will continue to be an important tool to combat the spread of infections until a vaccine is developed and distributed for any newly emerging variant or pathogen. While proving useful in slowing the spread of COVID-19 (Abaluck et al. 2022; Koelle et al. 2022) and other diseases (Mills et al. 2004), NPIs along with the vaccines themselves have been fraught with political controversy, leading to varying degrees of noncompliance. Given that vaccinations and NPIs are two main methods of stopping or slowing the spread of the disease, a Susceptible-Exposed-Asymptomatic-Infected-Recovered (SEAIR) model based on COVID-19 transmission was constructed to understand how these methods would work in concert to ease both hospital and societal burden.

During the rapid spread of COVID-19 infections early in the pandemic, hospital resources were being tested to their breaking point both in terms of physical resources (e.g., ventilator shortages) and staff (e.g., nursing shortages) (Nicola et al. 2020). Along with hospital burdens, there were also a number of societal burdens such as the increasing sense of isolation as the pandemic continued to financial burdens as businesses closed (Nicola et al. 2020). One way to lessen these burdens, which has been continually advocated, is to decrease the number of infected individuals both at a single point in time (as measured by peak infection) and across the course of the entire pandemic (as measured by overall number

of infections). The latter essentially flattens the curve of the epidemic where by the majority of infections occur over a longer period of time. By both lowering the disease burden and “flattening the curve,” one relieves stress on physical and staffing resources at hospitals (Feng et al. 2020) and gives clinicians time to improve intervention strategies (Koelle et al. 2022). At the beginning of a pandemic when vaccines may be unavailable, particularly for a newly emerging disease, the first line of defense is NPIs, which can be deployed relatively rapidly as other measures are in development (Glaubitx and Fu 2020). NPIs – such as wearing a mask or social distancing – can also be continually enforced after vaccines are available to lighten both societal and hospital burdens as individuals line up for vaccinations and, hopefully, the population as a whole approaches herd immunity.

In general, past mathematical modeling efforts have shown that mask wearing tends to lower transmission (Eikenberry et al. 2020) and decrease the disease burden associated with COVID-19 (Ngonghala et al. 2020) especially when combined with other NPI strategies (Ferguson et al. 2020). However, relying solely on NPIs may build up the number of susceptibles in a population, which could then lead to larger outbreaks in the future (Baker et al. 2020) if vaccines are ineffective or poorly distributed. Yet, most of these studies were conducted prior to the widespread availability of SARS-CoV2 vaccines. When combined with vaccination efforts, NPIs such as mask wearing may be particularly effective at lowering hospital and societal burdens by decreasing case loads while also allowing the remaining uninfected individuals to be vaccinated and, in turn, lessen the number of susceptibles in the population.

2 The Model

To examine the interplay between vaccination efforts and NPI usage, a Susceptible-Exposed-Asymptomatic-Infected-Recovered or SEAIR model (Eqs. 1–9) was constructed that separates individuals into two groups that either comply or do not comply with NPI mandates (Fig. 1). In particular, the model focuses on one of the primary NPIs used throughout the COVID-19 pandemic – the wearing of a mask to decrease transmission. The model equations are

$$\begin{aligned} \frac{dS}{dt} = & -\beta(I + (1 - \varepsilon)I_{NPI} + b_A A + \\ & (1 - \varepsilon)b_A A_{NPI})S - \rho\xi S, \end{aligned} \quad (1)$$

$$\begin{aligned} \frac{dS_{NPI}}{dt} = & -\beta(I + (1 - \varepsilon)I_{NPI} + b_A A + \\ & (1 - \varepsilon)b_A A_{NPI})(1 - \varepsilon)S_{NPI} - \rho_{NPI}\xi S_{NPI}, \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{dE}{dt} = & \beta(I + (1 - \varepsilon)I_{NPI} + b_A A + \\ & (1 - \varepsilon)b_A A_{NPI})S - \gamma_E E, \end{aligned} \quad (3)$$

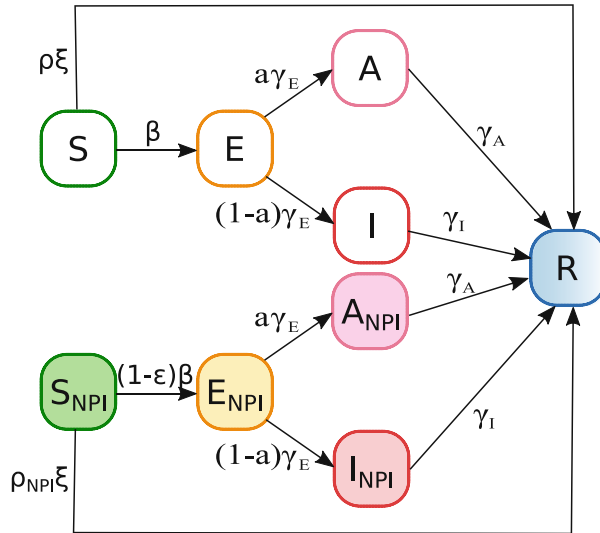


Fig. 1 Conceptual figure of SEAIR dynamics where individuals either comply with NPI requirements, which is signified by the NPI subscript in the state variables and the shaded boxes, or do not comply. State variables include susceptible (S), exposed (E), asymptomatic (A), infected (I), and recovered (R) and are placed in individual boxes. Recovered individuals are a mixture of individuals who comply and those that do not comply since they can no longer pass on the infection and are included as a single state. Arrows represent transitions from state to state and the parameters next to the arrow represent transition rates from state to state. The parameters in the model include β , disease transmission rate; γ_x , removal rate from state x ; a , proportion of asymptomatic individuals; ρ , vaccination rate for individuals not following NPI guidelines; ρ_{NPI} , vaccination rate for individuals following NPI guidelines; ξ , vaccine efficacy; and ε , NPI efficacy

$$\frac{dE_{NPI}}{dt} = \beta(I + (1 - \varepsilon)I_{NPI} + b_A A + (1 - \varepsilon)b_A A_{NPI})(1 - \varepsilon)S_{NPI} - \gamma_E E_{NPI}, \quad (4)$$

$$\frac{dA}{dt} = a\gamma_E E - \gamma_A A, \quad (5)$$

$$\frac{dA_{NPI}}{dt} = a\gamma_E E_{NPI} - \gamma_A A_{NPI}, \quad (6)$$

$$\frac{dI}{dt} = (1 - a)\gamma_E E - \gamma_I I, \quad (7)$$

$$\frac{dI_{NPI}}{dt} = (1 - a)\gamma_E E_{NPI} - \gamma_I I_{NPI}, \quad (8)$$

$$\frac{dR}{dt} = \gamma_A A + \gamma_A A_{NPI} + \gamma_I I + \gamma_I I_{NPI} + \rho\xi S + \rho_{NPI}\xi S_{NPI}. \quad (9)$$

Equations 1–9 divide individuals into nine separate states depending on whether they are (S)usceptible, (E)xposed, (A)symptomatic, (I)nfectious, or (R)ecovered. S, E, A, and I are further divided based on whether individuals follow NPI guidelines, denoted by the NPI subscript, or do not follow NPI guidelines, no subscript (Fig. 1). Individuals move from state to state based on assigned parameter values (Table 1). At the beginning of each simulated epidemic, 25 infected individuals were introduced into a population of 25,000 people. The population is divided into individuals who either comply or do not comply with NPI mandates. Both the fraction of individuals complying, α , and the vaccination ratio for individuals not practicing or practicing NPI (ρ/ρ_{NPI}) were varied. At the beginning of the epidemic, infected individuals were divided into either individuals practicing or not practicing NPI based on the value of α , the fraction complying.

For the analysis, four sets of model simulations are based on Eqs. 1–9. The first set consisted of setting all the parameters in the model (Eqs. 1–9) to their most likely value based on the literature (Table 1) while varying α and the vaccination ratio. For the second set, parameter values from the best fit normal distributions based on the literature (Table 1) were randomly chosen. Since parameters were chosen randomly, the simulations were run 200 times for each combination of α and the vaccination ratio. These simulations were used to establish confidence intervals for estimates of hospital and societal burden. The third set of models consisted of using a stochastic Markov chain Monte Carlo (MCMC) compartmental model based on Eqs. 1–9 with fixed parameter estimates (Table 1). For each combination of α and the vaccination ratio, the model was run 30 times. Initial runs for a limited set of parameters were conducted 50 and 100 times and showed no appreciable change in the results as compared to 30 runs of the model. Given the computational burden associated with the increased number of simulations, 30 runs per parameter combination were deemed sufficient. The stochastic differential equations based model explored

Table 1 Parameter values used in the SEAIR model (Fig. 1). For parameter values associated with transmission such as reproductive rate of spread, latency period, and asymptomatic/infectious period that were derived from the literature, the variability surrounding these estimates as indicated by the 95% confidence intervals (CI) in the parentheses was also taken into account. Given that the model tracks the number of individuals and that the asymptomatic and infectious period are the same, the transmission rate, $\beta = R_0\gamma_{A,I}$ (see Fig. 1 and Keeling and Rohani 2008.) Parameter values are based on estimates associated with the first wave of COVID-19

Parameter	Value(s)
R_0 , reproductive rate of spread (Davies et al. 2020)	2.7 (1.6, 3.9)
$1/\gamma_E$, latency period (Brauner et al. 2021)	5.79 (5.48, 6.11)
$1/\gamma_{A,I}$, asymptomatic/infectious period (Li et al. 2020)	3.45 (3.24, 3.70)
ε , NPI efficacy (Ngonghala et al. 2020)	0.50
ξ , vaccine efficacy	0.95
a , proportion asymptomatic	0.35
b_A , relative transmission of asymptomatic	1.0
base vaccination rate	0.002
ρ/ρ_{NPI} , vaccination ratio based on NPI	varies

how stochasticity influenced the infection process and its subsequent impacts on transmission dynamics. The fourth and final set consisted of randomly choosing parameter values from the best fit normal distributions (Table 1) for the stochastic MCMC version of the SEAIR model (Eqs. 1–9).

3 The Simulations

By examining a single outbreak using estimated parameters gathered from the literature (Table 1) and a set of ordinary differential equations (Eqs. 1–9, Fig. 1), it becomes apparent that NPI compliance decreases the overall burden associated with the disease as defined by the peak number of infected individuals and also increases the period of time over which the epidemic occurs (Fig. 2). The latter essentially flattens the curve (Feng et al. 2020). If one reasonably assumes that a certain portion of infected individuals will end up hospitalized, both of these factors, decreasing the peak number of infections and “flattening” the curve, decrease the hospital and societal burden associated with a single outbreak across a wide range of NPI compliance (Fig. 2). Notably, individuals who do not comply with NPI mandates represent the vast majority of individuals that contract COVID-19 (Fig. 2, solid red line) and, subsequently, spread the disease throughout the community. One way, in which, to decrease their impact on disease transmission would be to make these individuals the focus of vaccination campaigns.

To further examine how vaccination of compliant versus noncompliant individuals affects overall burden, which here was measured as the percent of the population that has recovered from being infected, both the degree of NPI compliance and the ratio of vaccines given to noncompliant versus NPI-compliant individuals were systematically varied. This effectively varies the degree of targeting used in vaccine

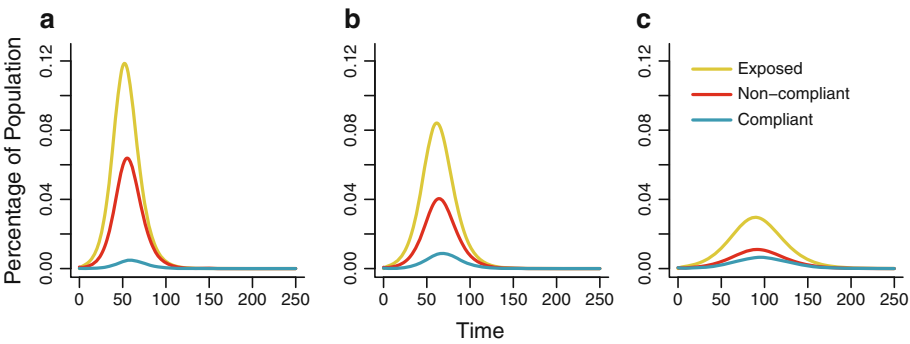


Fig. 2 Percent of population over time exposed (yellow line) and infected, whether not compliant (red line) or compliant (blue line) with NPIs given varying degrees of compliance – 10%, 25%, or 50% compliance from left to right (a, b, and c, respectively). The figure shows that the vast majority of infected individuals are ones who have not complied with NPI mandates. As compliance increases, overall infection rates decline, and the curve becomes flatter indicating that infections are spread out over a longer period of time, both of which would lessen hospital and societal burdens

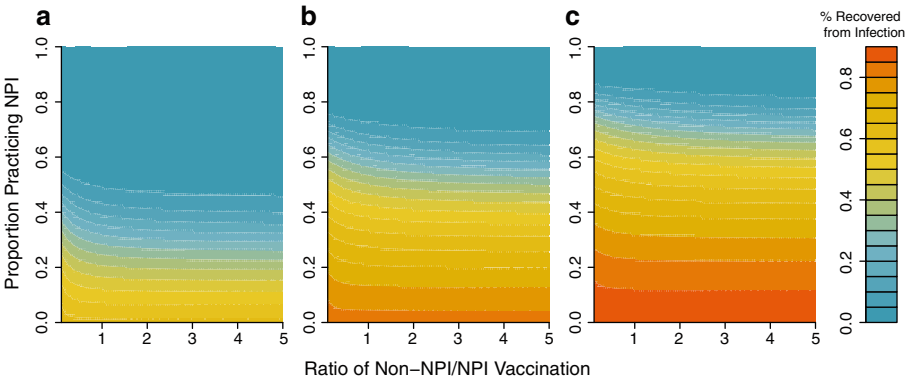


Fig. 3 A heat map of the intensity of the epidemic as measured by the percent of the population who have recovered from the disease while varying the percentage of the population practicing NPI, α , and the ratio of vaccinating noncompliant (non-NPI) to compliant (NPI) individuals, ρ/ρ_{NPI} . Warmer colors indicate a higher percentage of the population has become ill and recovered from the disease. The associated grids represent the 10% lower bound estimate (a), the 50% median estimate (b), and the 90% upper bound estimate (c) based on drawing parameter values from their respective distributions (Table 1). As NPI compliance drops and vaccination favors compliant individuals, the intensity of the epidemic increases. The median estimate corresponds to the deterministic model (Eqs. 1–9) with fixed parameters (Table 1). Note that recovered individuals consist of individuals who are no longer ill and those individuals who have died

campaigns. To do this, the SEAIR model (Eqs. 1–9) was run with randomly drawn parameter values from a distribution associated with each parameter derived from the literature (Table 1). When individuals in a population follow NPI guidelines, the overall burden drops dramatically (Fig. 3) from upward of 80–90% of the population on average contracting COVID-19 where relatively few individuals follow NPI guidelines to less than 10% of the population when the vast majority of individuals follow NPI guidelines. When lower percentages of individuals follow the guidelines and if the vast majority of individuals who get vaccinated are NPI compliant, disease burden is the highest (Fig. 3, lower left hand corner and Fig. 4). As vaccination efforts focus on individuals who chose not to follow NPI guidelines, the overall burden declines. However, simply following NPI guidelines by wearing a mask has a large effect. These results hold true when stochastic versions of these models were used while either keeping the transmission-associated parameters fixed or by varying the parameters (Fig. 5).

4 Discussion

Clearly, a combined strategy of NPIs and vaccination will decrease both hospital and societal burdens. If NPI mandates are followed, the use of NPIs clearly decreases the number of cases. However, mandates and compliance vary greatly.

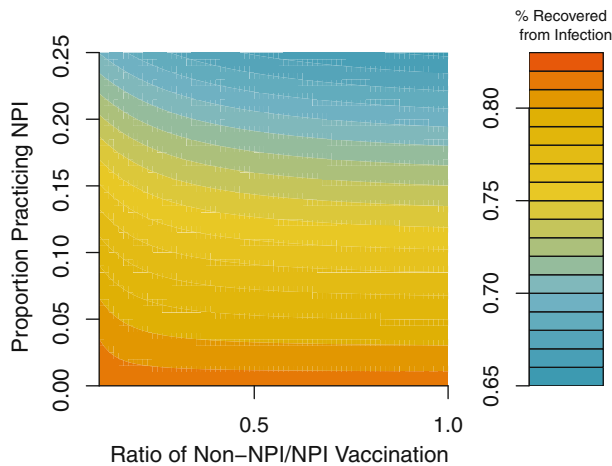


Fig. 4 A snapshot of the lower left corner of a heat map of the intensity of the epidemic for populations with low NPI compliance as measured by the percent of the population who have recovered from the disease while varying the ratio of vaccinating noncompliant (non-NPI) to compliant (NPI) individuals. Vaccination rates favor individuals who follow NPI guidelines. Warmer colors indicate a higher percentage of the population has become ill and recovered from the disease. The associated grid represents the 50% median estimate based on drawing parameter values from their respective distributions (Table 1). Note that recovered individuals consist of individuals who are no longer ill and those individuals who have died

For instance, in the United States, in over a quarter of the states, less than 30% of the population consistently wore masks in the middle of 2020 outbreak of SARS-CoV2 (Fischer et al. 2021). When considering ways to increase vaccine uptake or even NPI compliance, a one-size-fits-all approach is not necessarily the best way to move forward and a more targeted approach aimed, in particular, at the vaccine hesitant, and the NPI noncompliant will be more effective (Fig. 3). The question then becomes how best to target individuals that may be hesitant to get the vaccine, which are often the same individuals that do not wish to follow NPI guidelines (Bokemper et al. 2021). Vaccine mandates appear to be a major focus of making sure the populace is vaccinated such as when the United States required all healthcare and government workers to be vaccinated against COVID-19 (The White House 2021). However, enforced mechanisms undermine vaccine acceptance and decrease the general populace's trust that the vaccine is an effective measure to control outbreaks (Schmelz and Bowles 2021). Others advocate behavioral nudges whereby individuals are personally reminded to sign up for vaccination slots and that these nudges are phrased using ownership language (e.g., "your dose is ready for you") (Dai et al. 2021). Allowing individuals to choose a particular vaccine (e.g., Moderna vs. Pfizer COVID-19 vaccines) also gives the general public a sense of ownership or control (Hughes et al. 2021). Another approach is through prosocial messaging. Prosocial messaging, which highlights that vaccinations benefit society as a whole (e.g., herd immunity or protecting the immunocompromised), tends to decrease

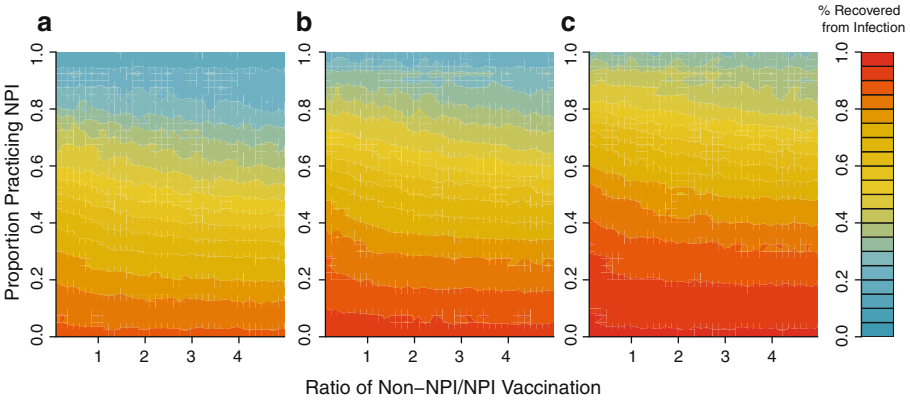


Fig. 5 A heat map of the intensity of the epidemic as measured by the percent of the population who have recovered from the disease while varying the percentage of the population practicing NPI, α , and the ratio of vaccinating noncompliant (non-NPI) to compliant (NPI) individuals, ρ/ρ_{NPI} . Warmer colors indicate a higher percentage of the population has become ill and recovered from the disease. The associated grids represent the 10% lower bound estimate (a), the 50% median estimate (b), and the 90% upper bound estimate (c) based on drawing parameter values from their respective distributions (Table 1). As NPI compliance drops and vaccination favors compliant individuals, the intensity of the epidemic increases. The median estimate corresponds to the deterministic model with fixed parameters. Note that recovered individuals consist of individuals who are no longer ill and those individuals who have died. Model simulations are based on the stochastic MCMC version of the SEIAR model (Eqs. 1–9) where parameters were randomly chosen. The median estimates presented here are comparable to the stochastic MCMC version where parameters were fixed. The latter is not shown

vaccine hesitancy (James et al. 2021). In the United States, white evangelicals tend to be individuals who are most prone to be anti-vaccine. Bokemper et al. (2021) found that messages associated with community interest along with individual embarrassment for not being vaccinated were the most effective means of increasing potential vaccination rates prior to widespread vaccine availability. However, once the vaccine became widely available, they found no message was effective in persuading individuals to become vaccinated (Bokemper et al. 2021). Thus, even targeted messaging may not be effective to the more recalcitrant individuals. But maybe it is not just the message that is important, but the messenger may also matter. When a trusted scientific source (i.e., Francis Collins, the former head of the US National Institute of Health (NIH)) advocated for vaccination to protestant groups and also mentioned his own faith, there was a decrease in vaccine hesitancy as compared to when he solely advocated for vaccination (Chu et al. 2021). If, instead, one wants to broadcast a general message, messages from nonpartisan experts or bipartisan coalitions will provide the largest effect (Flores et al. 2022). In the Czech Republic, when individuals became aware that the vast majority of doctors supported vaccination, vaccine uptake increased considerably. According to the authors, the lack of public knowledge surrounding doctors’ general opinion arose from “balanced” presentation of pro- and anti-vaxxer sentiment among doctors by

the news media (Bartoš et al. 2022). A controversial approach has been to offer financial incentives, which are thought to have unintended negative consequences of reducing healthy behaviors when incentives are not offered. However, a study of individuals in Sweden and the United States showed no unintended negative consequences associated with financial incentives for receiving the first dose of the COVID-19 vaccine (Schneider et al. 2023). Thus, a combined strategy of finding both the right message and the right messenger may be the most effective in increasing vaccine uptake for individuals who may not necessarily follow other recommended means of decreasing transmission of an emerging epidemic disease.

While there is always uncertainty associated with model parameters, different parameters associated with transmission provide different challenges. For instance, there may be greater uncertainty associated with the reproductive rate of spread of the disease, R_0 , given that R_0 estimates are often derived from case counts, as compared to other standard parameters such as the latency period or infectious period. These periods associated with transmission are often derived from clinical studies. Thus, estimates of latency and infectious period provide sound footing for any disease model. This approach of using clinical studies has been used in the past to estimate the uncertainty associated with transmission and, in particular, R_0 (Elder et al. 2006, 2013). For this study, estimates of R_0 were directly derived from outbreak dynamics (Davies et al. 2020). Given the continued evolution of SARS-CoV2, these estimates changed as more and more variants evolved and circulated (Ito et al. 2022). To account for this, additional simulations were also run where R_0 was varied. The focus on R_0 arose due to the higher uncertainty associated with this parameter. If either the mean value or the overall distribution (i.e., the variance) was changed, similar results emerge where following NPI guidelines and focusing vaccination efforts on those who fail to follow those guidelines results in less overall cases (Figs. 6 and 7). Thus, the overall conclusions are relatively robust for a wide range of R_0 . However, if R_0 approaches the reproductive rate of spread associated with highly contagious diseases such as measles with an $R_0 \approx 11$ to 16 (Anderson and May 1991), the infection spreads so rapidly through the population that both NPI compliance and vaccination efforts do little to lessen the spread with 95% to 98% of the population becoming infected (Fig. 6).

In the analysis, a relatively simple SEAIR model (Eqs. 1–9) was constructed. More complex models can be readily built on this framework (e.g., ones that incorporate age structure). Yet, more complex models require more parameter estimates. A number of the early models of COVID-19 transmission were rather complex. These models often fixed parameter estimates at a single value and did not allow for stochasticity in most parameter values (e.g., Ferguson et al. 2020). By fixing parameters, these models offer greater certainty in their model outcomes that may not be fully warranted (Elder et al. 2006, 2013). Thus, given the data, a simpler framework was chosen that readily allows for insight into how changing a number of parameter values associated with disease dynamics affects hospital and societal burdens.

Another simplification associated with the SEAIR model (Fig. 1) revolves around which NPI measure to consider when constructing the model. This modeling

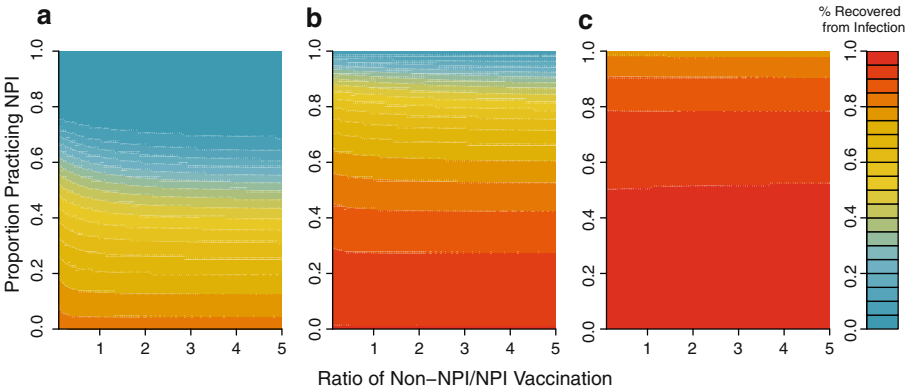


Fig. 6 The effects of varying R_0 on percent of the population recovered from the disease while varying the percentage of the population practicing NPI and the ratio of vaccinating noncompliant (non-NPI) to compliant (NPI) individuals. Warmer colors indicate a higher percentage of the population has become ill and recovered from the disease. The associated grids represent the 50% median estimate for $R_0 = 2.7$ (a), $R_0 = 5$ (b), and $R_0 = 10$ (c). R_0 values were drawn from a normal distribution with a standard deviation based on Davies et al. (2020). The other model parameter values are drawn from literature-based estimates (Table 1). The R_0 values used are based on estimates of the reproductive rate of spread of COVID-19 for the Alpha, Delta, and Omicron variant (Ito et al. 2022), respectively. As NPI compliance drops and vaccination favors compliant individuals, the intensity of the epidemic increases. The median estimate corresponds to the deterministic model (Eqs. 1–9) with fixed parameters (Table 1). Note that recovered individuals consist of individuals who are no longer ill and those individuals who have died

framework focused on mask wearing and not other NPI measures. Mask wearing essentially decreases the probability of transmission when contacted by an infected individual as opposed to other NPI strategies, such as social distancing, which decreases the number of contacts (Katul et al. 2020). However, the latter can also be easily incorporated into the modeling framework by altering transmission rates. Lastly, viral evolution over the course of an outbreak was not explicitly considered, which continued to be a concern during the COVID-19 pandemic given the rise in COVID-19 variants. These new variants can decrease the effectiveness of the vaccine and can change a number of SEAIR model parameters such as transmission rates (Ito et al. 2022; Liu and Rocklöv 2021). However, these changes can be easily incorporated into the model and, the simulation results show (Fig. 6), changing transmission rates does not qualitatively change the results.

5 Summary/Conclusion

In general, effective means of decreasing transmission after contact through NPI compliance lowers both the hospital and societal burden. When this is combined with prioritizing the vaccination of those individuals who chose not to follow NPI guidelines, the burden declines further. It is rewarding individuals who chose not to

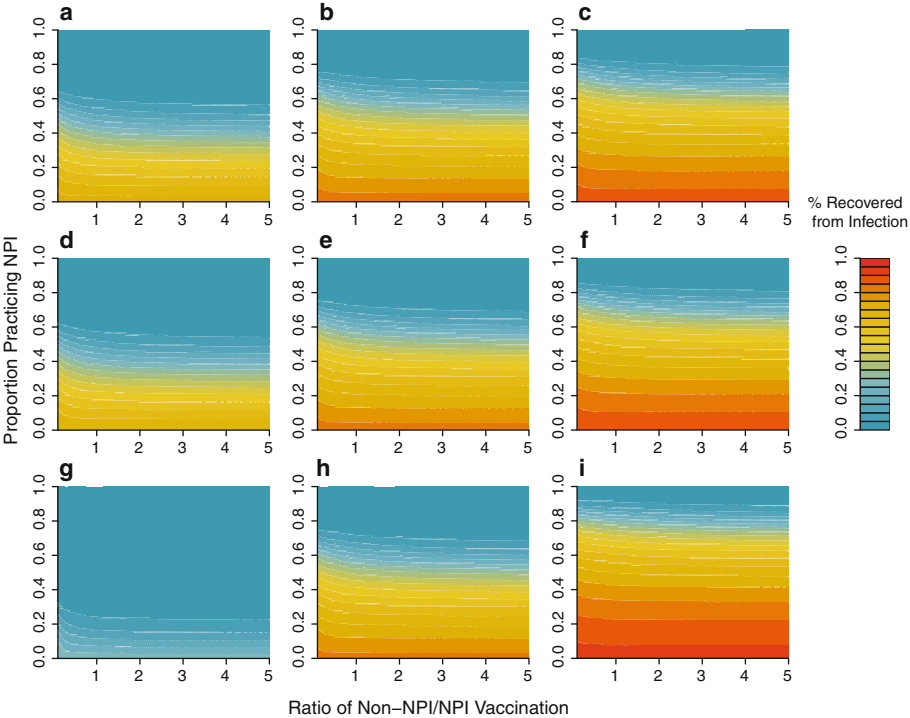


Fig. 7 The effects of increasing or decreasing the variance associated with R_0 on percent of the population recovered from the disease while varying the percentage of the population practicing NPI and the ratio of vaccinating noncompliant (non-NPI) to compliant (NPI) individuals. Warmer colors indicate a higher percentage of the population has become ill and recovered from the disease. For each of the simulations $R_0 = 2.7$ (Davies et al. 2020) and the variation about R_0 is either decreased by half or doubled given literature derived estimates of R_0 (Davies et al. 2020). The associated grids represent the 10% lower bound, 50% median, and the 90% upper bound for one half the variance (a, b, and c, respectively), the original estimate of the variance (d, e, and f, respectively), and double the variance (g, h, and i, respectively). The other model parameter values are drawn from literature-based estimates (Table 1). As NPI compliance drops and vaccination favors compliant individuals, the intensity of the epidemic increases. Note that recovered individuals consist of individuals who are no longer ill and those individuals who have died

follow the rules since those individuals reap the benefits of others practicing NPI and are essentially free riders. When faced with a global pandemic, the overall societal benefit associated with rewarding noncompliers clearly outweighs the rewarding of bad actors or free riders in the system. To assure that free riders do not take advantage of the system, the healthcare system has often relied on vaccine mandates. The historical effectiveness of vaccine mandates is well noted (e.g., smallpox and polio) (Malone and Hinman 2003; Elder et al. 2006). In essence, mandates guard against the tragedy of the commons whereby the individual who wishes not to be vaccinated does not get an opportunity to threaten the community’s herd immunity

(Malone and Hinman 2003). However, mandates may not work in future pandemics given that other mandates for childhood vaccinations have not entirely stamped out early childhood disease such as measles (Jansen et al. 2003). Therefore, society needs to move toward a more nuanced message in order to prevent the tragedy of the commons and protect the common good that is public health.

Data Availability

The code used to produce the results in this manuscript is available at <https://github.com/elderd-lab/COVID19-NPI-Vaccination>.

6 Cross-Referencenes

- ▶ [Data-Driven Approach to Analysis of SIR \(Susceptible-Infected-Removed/Recovered\)-Type Models: The Principle of Parsimony Applied to Epidemics Modeling in the Age of COVID-19](#)
- ▶ [Dynamical Survival Analysis for Epidemic Modeling](#)
- ▶ [Markov Decision Processes and Best Responses of Epidemic Models](#)
- ▶ [Mathematical Modeling and Wastewater-Based Epidemiology](#)
- ▶ [MONOPOLI: A Customizable Model for Forecasting COVID-19 Around the World Using Alternative Nonpharmaceutical Intervention Policy Scenarios, Human Movement Data, and Regional Demographics](#)
- ▶ [Retrospective Modeling of the Impact of Vaccination and Non-pharmaceutical Interventions on the COVID-19 Pandemic](#)

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